

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The tcpdump logs indicate that port 53 is unreachable when attempting to request the IP address for the domain as no service was listening on the receiving DNS port(since port 53 is for DNS service). The first two lines of the log show the outgoing UDP message from our computer to the DNS server to retrieve the IP address for the domain "www.yummyrecipesforme.com". The third and fourth lines of the log show incoming ICMP error messages which shows that port 53 is unreachable. As a result the IP address could not be retrieved and the website could not be accessed.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred at 1:24 pm when the customers of the client reported the issue. The network security team responded and ran tests with the network protocol analyzer tool. The logs revealed that the udp packet could not be delivered to port 53, which is the port for DNS service. An error, "destination port unreachable", was seen when attempting to access the company's website. This is because our request for the IP address for the domain "www.yummyrecipesforme.com" does not go through to the DNS server. Our next steps include checking if our firewall is blocking the port. If not, it might be a DoS attack which caused the DNS server to crash.